

Navigation in GNSS-Denied Environments Using MEMS-Grade Sensors and Geophysical Anomalies: A Particle Filter Approach

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Outline

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- 2 Methodology
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GNSS Is Ubiquitous—and Vulnerable

GNSS limitations:

- Relies on weak RF signals from orbit
- **Denial:** urban canyons, indoors, underground, underwater
- **Jamming:** electronic warfare can block signals over wide areas
- **Spoofing:** false signals inject position errors silently

Consequences:

- Inertial Navigation Systems (INS) drift rapidly without GNSS corrections
- Critical for aviation, maritime, autonomous vehicles, military

Rapid INS Drift

Without corrections, MEMS-grade IMUs accumulate **kilometers of error** over minutes of flight/drive time.

Alt-PNT is Needed

Alternative Positioning, Navigation, and Timing (Alt-PNT) that does **not** rely on external RF signals.

Geophysical Navigation: A Passive Alternative

Key idea: Earth's gravity and magnetic fields have spatial structure.

- **Gravity anomaly:** deviation of local gravity from theoretical model—mapped globally at ~ 1.8 km resolution (IGPP)
- **Magnetic anomaly:** deviation of local field from IGRF model—mapped at ~ 5.6 km resolution (WDMAM)
- Compare *sensed* values to *map* values $\rightarrow$ position estimate via map-matching
- Entirely passive, no RF emissions

Prior art: demonstrated in aerospace/maritime with *navigation-grade* sensors (RLG/FOG IMUs, dedicated gravimeters)

Open Question

Does geophysical navigation remain viable with **MEMS-grade sensors** (\$10–\$1,000) and publicly available, lower-resolution maps?

Why MEMS?

Smartphones, drones, small autonomous vehicles—the platforms most exposed to GNSS denial are the ones that cannot carry high-grade sensors.

Prior Work and This Contribution

Prior work (Brodovsky & Dames, ION 2025):

- Integrated gravity & magnetic anomaly into an **Unscented Kalman Filter** (UKF) INS
- MEMS-grade sensors + open-source maps
- Demonstrated feasibility; limited by Gaussian assumption

Problem with UKF in map-matching:

- Anomaly map may produce *multimodal* posteriors (same reading, multiple locations)
- UKF collapses to single Gaussian—can diverge

This Work

Replace UKF with a **Rao-Blackwellized Particle Filter** (RBPF):

- Particles represent non-Gaussian, multimodal position hypotheses
- Kalman filter per particle for linear states
- Enables efficient representation of complex posteriors

First application of RBPF geophysical navigation to the MEMS-grade sensor regime.

Dataset and Experimental Setup

MEMS-Nav Dataset:

- Smartphone-grade sensors via *Sensor Logger* app
- 21 trajectories, mostly highway driving
- Duration: 30 min – several hours
- Distance: tens to hundreds of km
- Sensors: IMU (accel + gyro), GNSS, magnetometer, barometer

Geophysical Maps:

- Gravity: IGPP free-air anomaly (~ 1.8 km resolution)
- Magnetic: WDMAM (~ 5.6 km resolution)
- Interpolated via GMT bilinear interpolation

Controlled Comparison

Identical dataset, degradation scenarios, and maps as prior UKF work.

Only the **filter architecture** changes (RBPF vs. UKF).

Three Test Configurations

- 1 Gravity anomaly aiding only
- 2 Magnetic anomaly aiding only
- 3 Combined gravity + magnetic

Plus two baselines: RBPF (no geo) and UKF (no geo)

GNSS Degradation Model

Simulating realistic GNSS degradation:

- Not simple white noise—uses **first-order autoregressive (AR(1))** error injection
- **Position errors:** $\rho_{\text{pos}} = 0.99$, $\sigma_{\text{pos}} = 5 \text{ m}$ (slowly drifting bias)
- **Velocity errors:** $\rho_{\text{vel}} = 0.95$, $\sigma_{\text{vel}} = 5 \text{ m/s}$
- Covariance **inflated** $\times 15$: forces filter to distrust GNSS

Represents:

- Urban canyons, multipath
- Moderate jamming conditions
- Receiver reporting degraded accuracy

Why AR(1)?

Real GNSS errors under multipath or jamming are **temporally correlated**, not i.i.d.

AR(1) creates slowly wandering biases that are more challenging for the filter than white noise.

Same Degradation Applied to All

Identical parameters for both baseline and geophysically-aided runs $\Rightarrow$ controlled comparison.

Rao-Blackwellized Particle Filter Architecture

Key idea—state partitioning:

$$\mathbf{x}_t = \begin{bmatrix} \underbrace{\mathbf{x}_t^n}_{\text{nonlinear}} \\ \underbrace{\mathbf{x}_t^l}_{\text{linear}} \end{bmatrix}$$

Nonlinear states (particles):

$$\mathbf{x}^n = \begin{bmatrix} L \\ \lambda \end{bmatrix} \quad (\text{lat, lon})$$

Particles explicitly represent multimodal position hypotheses.

Linear states (Kalman filter per particle):

$$\mathbf{x}^l = [h, v_n, v_e, v_d, \phi, \theta, \psi, \mathbf{b}_a, \mathbf{b}_g, b_{geo}]^\top$$

Why This Works

Map-matching is **highly nonlinear** in (lat, lon): same anomaly reading can match many map locations.

Altitude, velocity, attitude, and biases are approximately **linear given position**—handled exactly by KF.

Result: far fewer particles needed vs. full 16-D particle filter.

RBPF: Prediction, Update, and Resampling

Prediction (each particle):

- Propagate position via strapdown equations
- KF propagates linear states (velocity, attitude, biases)

Measurement update (geophysical):

$$\hat{\mathbf{z}}^{(i)} = \underbrace{M(\mathbf{x}^{n,(i)})}_{\text{map lookup}} + \underbrace{\mathbf{H} \mathbf{x}^{l,(i)}}_{\text{geo bias}}$$

$$w^{(i)} \propto w_{\text{prev}}^{(i)} \cdot \mathcal{N}(\nu^{(i)}; 0, \mathbf{S}^{(i)})$$

Each particle's KF also updates its linear states.

Resampling:

- Monitor effective sample size:

$$N_{\text{eff}} = \frac{1}{\sum_i (\tilde{w}^{(i)})^2}$$

- When $N_{\text{eff}} < N_{\text{thresh}}$: **systematic resampling**
- Eliminates low-weight particles; duplicates high-weight ones

State estimate:

$$\hat{\mathbf{x}}_t = \sum_i \tilde{w}_t^{(i)} \mathbf{x}_t^{(i)}$$

Weighted mean across all particles.

Geophysical Measurement Model

Measurement model:

$$z_{\text{geo}} = M(L, \lambda) + b_{\text{geo}} + \nu_{\text{geo}}$$

- $M(L, \lambda)$: bilinear map interpolation at particle hypothesis
- b_{geo} : slowly varying sensor bias (estimated by KF)
- ν_{geo} : Gaussian measurement noise

Anomaly residuals (from MEMS-Nav dataset):

Type	Mean	Std. Dev.
Gravity	5.97 mGal	48.93 mGal
Magnetic	40,364 nT	41,303 nT

Processing Pipeline

- 1 Subtract theoretical model from sensor reading $\rightarrow$ anomaly residual
- 2 Compare residual to anomaly map value at each particle's (lat, lon)
- 3 Update particle weight based on how well they agree

Both Sources

Gravity and magnetic anomalies are processed **independently**, each with its own bias state and noise parameters.

Results Overview: 21 Trajectories

RBPF (geo-aided) vs. RBPF baseline:

Aiding	Mean RMSE Δ	Median
Gravity	−1,565 km	−1,476 km
Magnetic	−1,565 km	−1,475 km
Combined	−1,566 km	−1,476 km

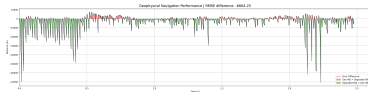
- **All 21 trajectories** show improvement vs. RBPF baseline
- Confirms: geophysical aiding, not filter architecture, drives the gain
- GNSS-only RBPF drifts into the thousands of km with low-quality GNSS

RBPF (geo-aided) vs. UKF baseline:

Aiding	Mean RMSE Δ	Median
Gravity	−234 m	−1,177 m
Magnetic	−520 m	−496 m
Combined	−1,395 m	−659 m

- More modest gains vs. UKF (0.2–1.4 km mean RMSE)
- UKF degrades deterministically; RBPF has stochastic drift
- Combined aiding shows best RBPF advantage

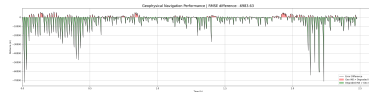
Example: Strong Performance



(a) Magnetic aiding



(b) Gravity aiding



(c) Combined aiding

Figure: Trajectory 2023-08-06: geophysically-aided RBPf (orange) vs. RBPf baseline (blue) and UKF baseline (green). All three modalities show substantial, consistent improvement.

Example: Weak Performance



(a) Magnetic aiding



(b) Gravity aiding

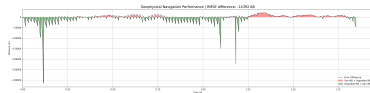


(c) Combined aiding

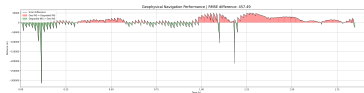
Figure: Trajectory 2025-06-18: minimal improvement over UKF baseline. Performance is dependent on favorable anomaly field characteristics along the trajectory.

Geophysical aiding efficacy depends on anomaly field variability, trajectory geometry, and sensor noise—not all regions offer strong observability.

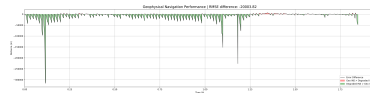
Example: Combined Aiding Synergy



(a) Magnetic only



(b) Gravity only



(c) Combined

Figure: Trajectory 2025-07-08: combined anomaly aiding yields substantial improvement over either single modality. Gravity and magnetic anomalies resolve ambiguities that neither can address alone.

The RBPF's particle representation enables genuine fusion of multiple non-Gaussian aiding sources—a key advantage over single-Gaussian filter architectures.

Main findings:

- ① **Geophysical aiding works:** Mean RMSE reduction of $\sim 1,562$ km vs. GNSS-only RBPF baseline across all 21 trajectories
- ② **Architecture matters modestly:** RBPF provides incremental 0.2–1.4 km RMSE improvement vs. UKF, with combined aiding showing the greatest benefit
- ③ **Synergistic fusion:** Simultaneous gravity + magnetic aiding can resolve ambiguities neither source can handle alone
- ④ **Trajectory-dependent:** Performance relies on favorable anomaly field characteristics; not all regions provide strong observability

Key Takeaway

The RBPF is a viable architecture for MEMS-grade geophysical navigation. It handles non-Gaussian posteriors from map-matching without the $O(N^{16})$ cost of a full particle filter—but the primary driver of accuracy remains the **information content of the anomaly measurement**, not the filter alone.

Current limitations:

- Simple Gaussian measurement model—ignores:
 - Directional dependence of scalar magnetometer
 - Altitude corrections, temporal variation
 - Platform magnetic signature
- Weighted-mean state estimate can lie between particle clusters in multimodal cases
- Performance is field-dependent with no online observability estimate

Future directions:

- ➊ **Physics-informed models:** Canciani airborne magnetic navigation model accounting for vector-scalar relationships and platform signatures
- ➋ **Multi-modal state estimation:** K-means, GMM, or DBSCAN clustering to handle distinct particle clusters before forming navigation solution
- ➌ **Integrity monitoring:** Online observability metrics to identify when geophysical aiding is trustworthy
- ➍ **Adaptive particles:** Concentrate particles where uncertainty is highest

Geophysical anomaly navigation with MEMS-grade sensors using a Rao-Blackwellized Particle Filter is viable.

What we showed

- RBPF successfully fuses gravity & magnetic anomalies with degraded GNSS and low-cost INS
- Consistent improvement across 21 diverse trajectories
- Combined aiding provides synergistic benefit

Open-source tools

strapdown-rs: Rust library for high-performance INS algorithms

RBPF implementation, GNSS degradation simulation, geophysical map integration—all open source.

Thank you! Questions?

Extended state vector for geophysically-aided RBPF:

$$\mathbf{x} = \left[L \quad \lambda \quad h \quad v_n \quad v_e \quad v_d \quad \phi \quad \theta \quad \psi \quad b_{a_x} \quad b_{a_y} \quad b_{a_z} \quad b_{g_x} \quad b_{g_y} \quad b_{g_z} \quad \mathbf{b}_{\text{geo}}^{\top} \right]^{\top}$$

- 16 states for single-anomaly aiding (gravity or magnetic)
- 17 states for dual-anomaly aiding (both)
- Nonlinear partition: (L, λ) — handled by particles
- Linear partition: all remaining 14–15 states — handled by Kalman filter per particle

Backup: Strapdown Mechanization

Attitude update (Z-Y-X Euler, body to nav frame):

$$\mathbf{C}_{t+1} \approx \mathbf{C}_t (I + \boldsymbol{\Omega}_b \Delta t) - (\boldsymbol{\Omega}_i + \boldsymbol{\Omega}_n) \mathbf{C}_t \Delta t$$

Velocity update:

$$\mathbf{v}_{t+1} \approx \mathbf{v}_t + \left(\hat{\mathbf{f}}_t + \mathbf{g}_t - (2\boldsymbol{\Omega}_i + \boldsymbol{\Omega}_n) \mathbf{v}_t \right) \Delta t$$

Position update (WGS-84):

$$L_{t+1} = L_t + \frac{1}{2} \left(\frac{v_{n,t}}{R_n(L_t) + h_t} + \frac{v_{n,t+1}}{R_n(L_{t+1}) + h_{t+1}} \right) \Delta t$$

$$\lambda_{t+1} = \lambda_t + \frac{1}{2} \left(\frac{v_{e,t}}{(R_e(L_t) + h_t) \cos L_t} + \frac{v_{e,t+1}}{(R_e(L_{t+1}) + h_{t+1}) \cos L_{t+1}} \right) \Delta t$$

Reference: Groves, *Principles of GNSS, Inertial, and Multisensor Integrated Navigation Systems*, 2nd ed., Ch. 5.3–5.5